Theoretical Questions - Q1

(a)

$$J(b_k, a_{k:k+L-1}) = \mathbb{E}_{z_{k+1:k+L}}\left[\sum_{l=0}^{L-1}\rho(b_{k+l}, a_{k+l}) + \rho_T(b_{k+L})\right] \tag{1}$$

Where the expectation is over the observations, with the assumptions we saw in the lecture.

(b)

We first separate the current time step from the Σ:

$$\begin{aligned}
J(b_k, a_{k:k+L-1}) &= \mathbb{E}_{z_{k+1:k+L}}\left[\sum_{l=0}^{L-1}\rho(b_{k+l}, a_{k+l}) + \rho_T(b_{k+L})\right] \\
&= \mathbb{E}_{z_{k+1:k+L}}\left[\rho(b_k, a_k) + \sum_{l=1}^{L-1}\rho(b_{k+l}, a_{k+l}) + \rho_T(b_{k+L})\right] \\
&= \rho(b_k, a_k) + \mathbb{E}_{z_{k+1:k+L}}\left[\sum_{l=1}^{L-1}\rho(b_{k+l}, a_{k+l}) + \rho_T(b_{k+L})\right]
\end{aligned}$$

Since $\rho(b_k, a_k)$ is deterministic at time k, we moved it outside. Then, we apply the chain rule:

$$\mathbb{E}_{z_{k+1:k+L}}[\cdot] = \mathbb{E}_{z_{k+1}}\left[\mathbb{E}_{z_{k+2:k+L}|z_{k+1}}[\cdot]\right]$$

Applying this to our objective function:

$$J(b_k, a_{k:k+L-1}) = \rho(b_k, a_k) + \mathbb{E}_{z_{k+1}}\left[\mathbb{E}_{z_{k+2:k+L}|z_{k+1}}\left[\sum_{l=1}^{L-1}\rho(b_{k+l}, a_{k+l}) + \rho_T(b_{k+L})\right]\right]$$

We notice that the term in the outer expectation is exactly the objective function starting from step $k+1$, until $k+L$. Thus, we can substitute it in:

$$J(b_k, a_{k:k+L-1}) = \rho(b_k, a_k) + \mathbb{E}_{z_{k+1}}\left[J(b_{k+1}, a_{k+1:k+L-1})\right] \tag{2}$$

Here, the observation z_{k+1} is calculated using the given observation model, and the belief b_{k+1} is available for each fixed observation and by applying the motion model.

(c)

$$V_k^{\pi}(b_k) = \mathbb{E}_{z_{k+1:k+L}}\left[\sum_{l=0}^{L-1}\rho(b_{k+l}, \pi(b_{k+l})) + \rho_T(b_{k+L})\right] \tag{3}$$

(d)

$$\begin{aligned} V_k^\pi(b_k) &= \mathbb{E}_{z_{k+1:k+L}} \left[\sum_{l=0}^{L-1} \rho(b_{k+l}, \pi(b_{k+l})) + \rho_T(b_{k+L}) \right] \\ &= \mathbb{E}_{z_{k+1:k+L}} \left[\rho(b_k, \pi(b_k)) + \sum_{l=1}^{L-1} \rho(b_{k+l}, \pi(b_{k+l})) + \rho_T(b_{k+L}) \right] \end{aligned}$$

The immediate reward $\rho(b_k, \pi(b_k))$ is deterministic given b_k. Again, using the chain rule:

$$V_k^\pi(b_k) = \rho(b_k, \pi(b_k)) + \mathbb{E}_{z_{k+1}} \left[\mathbb{E}_{z_{k+2:k+L}|z_{k+1}} \left[\sum_{l=1}^{L-1} \rho(b_{k+l}, \pi(b_{k+l})) + \rho_T(b_{k+L}) \right] \right]$$

We notice the term as the value function at time $k+1$:

$$V_k^\pi(b_k) = \rho(b_k, \pi(b_k)) + \mathbb{E}_{z_{k+1}} \left[V_{k+1}^\pi(b_{k+1}) \right] \tag{4}$$

(e)

The difference between the objective function $J(b_k, a_{k:k+L-1})$ and the value function $V_k^\pi(b_k)$ is inherently the difference between the open-loop and closed-loop settings.

$J(b_k, a_{k:k+L-1})$ represents the open-loop setting, where an action sequence $a_{k:k+L-1}$ is chosen in advance, and these actions are performed, regardless of the observations or the propagated belief.

On the other hand, $V_k^\pi(b_k)$ represents a closed-loop setting, where the policy π chooses an action on each timestep based on the current belief, which is affected by the observation.

Since $V_k^\pi(b_k)$ can adapt to the observations from the environment, the optimal value should be greater than or equal to the optimal $J(b_k, a_{k:k+L-1})$, as we saw in the lecture.

(f)

Plugging in our derivation from (b), we have:

$$\max_{a_{k:k+L-1}} J(b_k, a_{k:k+L-1}) = \max_{a_{k:k+L-1}} \left[\rho(b_k, a_k) + \mathbb{E}_{z_{k+1}} \left[J(b_{k+1}, a_{k+1:k+L-1}) \right] \right]$$

We split the max into the max over the current action a_k and the remaining sequence $a_{k+1:k+L-1}$. Since $\rho(b_k, a_k)$ doesn't depend on future actions, we have:

$$J_k^*(b_k) = \max_{a_k} \left[\rho(b_k, a_k) + \max_{a_{k+1:k+L-1}} \mathbb{E}_{z_{k+1}} \left[J(b_{k+1}, a_{k+1:k+L-1}) \right] \right] \tag{5}$$